Linear Algebra and Its Applications

Section 2.7: Applications to Computer Graphics

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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2.7 Applications to computer graphics

For two-dimensional graphics, write a point as $(x, y, 1)^T$. An affine transformation is then represented by

$$H = \begin{bmatrix} a & b & h \\ c & d & k \\ 0 & 0 & 1 \end{bmatrix}, \qquad H \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} ax + by + h \\ cx + dy + k \\ 1 \end{bmatrix}.$$

The upper-left block controls the linear part and the last column controls translation. A positive angle denotes counterclockwise rotation in the usual oriented plane. To perform transformation F first and G second, multiply their matrices as GF: the first action is the rightmost factor. We use $R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ and write $T_p = \begin{bmatrix} I_2 & p \\ 0^T & 1 \end{bmatrix}$ for translation.

Practice problem 2.7.P1: Rotate about a point other than the origin

Let $p = (-2, 6)^T$ and $\theta = -30°$. Subtract p to move the center to zero, rotate, then add p back. Algebraically this is $x \mapsto R_\theta(x - p) + p = R_\theta x + (p - R_\theta p)$. With $c = \sqrt{3}/2$, $s = -1/2$, the matrix is

$$T_p \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix} T_{-p} = \boxed{\begin{bmatrix} \sqrt{3}/2 & 1/2 & \sqrt{3} - 5 \\ -1/2 & \sqrt{3}/2 & 5 - 3\sqrt{3} \\ 0 & 0 & 1 \end{bmatrix}}.$$

For example $R_\theta p = (-\sqrt{3} + 3, 1 + 3\sqrt{3})^T$, which explains the translation column. Substituting $x = p$ gives p again, as a rotation center must be fixed.

Exercise 2.7.1: Embed the shear into homogeneous coordinates

Example 2 uses $(x, y) \mapsto (x + y/4, y)$. Preserve the final coordinate 1:

$$\boxed{\begin{bmatrix} 1 & 1/4 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}.$$

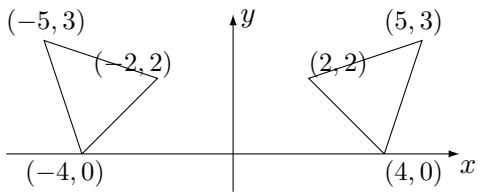
Multiplication by $(x, y, 1)^T$ gives $(x + y/4, y, 1)^T$, so the extra coordinate introduces no translation and does not change the original shear.

Exercise 2.7.2: Reflect and sketch a triangle

Reflection through the y-axis negates x and preserves y. Thus

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 2 & 5 \\ 0 & 2 & 3 \end{bmatrix} = \begin{bmatrix} -4 & -2 & -5 \\ 0 & 2 & 3 \end{bmatrix}.$$

The original vertices $(4, 0), (2, 2), (5, 3)$ therefore become $(-4, 0), (-2, 2), (-5, 3)$. Connect corresponding triples of vertices; reflection preserves straight segments and lengths.



Exercise 2.7.3: Translate first, rotate second

Translation by $(2,1)$ takes (x,y) to $(x+2,y+1)$. A 90° rotation then gives $(-(y+1),x+2)$. Reading off coefficients yields

$$R_{90^\circ}T_{(2,1)} = \boxed{\begin{bmatrix} 0 & -1 & -1 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}},$$

where the rotation matrix is understood in its homogeneous 3×3 form. The translated origin goes to $(-1,2)$, checking the last column.

Exercise 2.7.4: Translate before unequal scaling

The first action gives $(x-1,y+4)$, and the second gives $((x-1)/2, 3(y+4)/2)$. Therefore the matrix is

$$\boxed{\begin{bmatrix} 1/2 & 0 & -1/2 \\ 0 & 3/2 & 6 \\ 0 & 0 & 1 \end{bmatrix}}.$$

The translation column is scaled too, because the scaling is performed *after* translation.

Exercise 2.7.5: Reflect, then rotate

Reflection in the x-axis sends (x,y) to $(x,-y)$. Put $a=\sqrt{2}/2$. Rotation by 45° gives $(ax+ay, ax-ay)$, so

$$\boxed{\begin{bmatrix} a & a & 0 \\ a & -a & 0 \\ 0 & 0 & 1 \end{bmatrix}}.$$

Equivalently its upper-left block is $R_{45^\circ}\,\mathrm{diag}(1,-1)$.

Exercise 2.7.6: Reverse the preceding order

First rotation gives $(ax-ay, ax+ay)$. Reflecting through the x-axis now gives $(ax-ay, -ax-ay)$. Thus the matrix is

$$\boxed{\begin{bmatrix} a & -a & 0 \\ -a & -a & 0 \\ 0 & 0 & 1 \end{bmatrix}}, \qquad a=\sqrt{2}/2.$$

It differs from Exercise 5, illustrating why the order cannot be ignored.

Exercise 2.7.7: Rotate sixty degrees around a fixed point

Use the general formula $x \mapsto R_\theta x + p - R_\theta p$ with $p = (6,8)^T$, $\theta = 60°$. Since $R_\theta p = (3-4\sqrt{3}, 3\sqrt{3}+4)^T$, the result is

$$\boxed{\begin{bmatrix} 1/2 & -\sqrt{3}/2 & 3+4\sqrt{3} \\ \sqrt{3}/2 & 1/2 & 4-3\sqrt{3} \\ 0 & 0 & 1 \end{bmatrix}}.$$

Its action fixes $(6,8)$, which independently checks the translation terms.

Exercise 2.7.8: Rotate forty-five degrees around another point

Now $p = (3,7)^T$, and $R_{45°}p = (-2\sqrt{2}, 5\sqrt{2})^T$. Therefore

$$\boxed{\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 & 3+2\sqrt{2} \\ \sqrt{2}/2 & \sqrt{2}/2 & 7-5\sqrt{2} \\ 0 & 0 & 1 \end{bmatrix}}.$$

This is $T_p R_{45°} T_{-p}$, not $R_{45°} T_p$.

Exercise 2.7.9: Precompute a composite transformation

For $A(BD)$, the 2×100 product BD has 200 entries, each requiring two multiplications: 400 multiplications. Multiplication by A costs another 400, for 800 total. For $(AB)D$, forming the four entries of AB costs $4 \cdot 2 = 8$, and multiplying its result by D costs 400, for 408 total. Thus precomputing the 2×2 composite saves 392 multiplications. The advantage grows when the same transformations are applied to more vertices: combine the small transformation matrices once, then transform all data points with their product.

Exercise 2.7.10: Which geometric transformations commute?

Let $D(x) = \lambda x$, $R(x) = Rx$, and $T(x) = x + p$. Then $D(Rx) = \lambda Rx = R(\lambda x)$, so a uniform dilation and an origin-centered rotation always commute. By contrast,

$$D(Tx) = \lambda x + \lambda p, \qquad T(Dx) = \lambda x + p.$$

They commute exactly when $(\lambda - 1)p = 0$, and generally do not. Likewise $R(Tx) = Rx + Rp$ and $T(Rx) = Rx + p$, so rotation and translation commute exactly when $Rp = p$. A nontrivial planar rotation fixes only the zero vector, making a nonzero translation fail this test. For concrete failures take $p = (1,0)$, $\lambda = 2$, and a 90° rotation, and evaluate the two orders at $x = 0$.

Exercise 2.7.11: Two shear-and-scale factors produce a rotation

The first action is A_1, the second is A_2, so calculate $A_2 A_1$. Its upper-left block is

$$\begin{bmatrix} \sec\varphi & -\tan\varphi \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \sin\varphi & \cos\varphi \end{bmatrix} = \begin{bmatrix} \sec\varphi - \tan\varphi \sin\varphi & -\tan\varphi\cos\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix} = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix}.$$

The first simplification uses $(1-\sin^2\varphi)/\cos\varphi = \cos\varphi$. The last homogeneous row and column remain $(0,0,1)$, so this is rotation by φ. The stated factors require $\cos\varphi \neq 0$; the rotation itself still exists at other angles, but these particular factors do not.

Exercise 2.7.12: Three shears can also implement rotation

Write $t = \tan(\varphi/2)$, $s = \sin\varphi$. Ignoring the unchanged homogeneous coordinate for a moment, multiply the three 2×2 blocks:

$$\begin{bmatrix} 1 & -t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ s & 1 \end{bmatrix} \begin{bmatrix} 1 & -t \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1-ts & -2t+t^2 s \\ s & 1-st \end{bmatrix}.$$

Half-angle identities give $st = 1-\cos\varphi$ and $-2t+t^2 s = -\sin\varphi$, so the result is R_φ. For instance $\sin\varphi = 2t/(1+t^2)$ makes the second identity $-2t+2t^3/(1+t^2) = -2t/(1+t^2)$. Appending the homogeneous row and column yields precisely the claimed 3×3 rotation matrix. Each shear needs one nontrivial multiplication per point, whereas the usual rotation needs four. The factorization requires $\tan(\varphi/2)$ to be defined, excluding odd multiples of $180°$.

Exercise 2.7.13: Factor a general affine transformation

Block multiplication gives

$$\begin{bmatrix} A & p \\ 0^T & 1 \end{bmatrix} = \begin{bmatrix} I_2 & p \\ 0^T & 1 \end{bmatrix} \begin{bmatrix} A & 0 \\ 0^T & 1 \end{bmatrix}.$$

The right factor acts first and sends $(x,1)$ to $(Ax,1)$. The left then adds p, giving $(Ax+p,1)$. Thus the transformation is a linear transformation of the plane followed by a translation, as requested. No inverse or invertibility of A is needed.

Exercise 2.7.14: Separate the linear rotation from its final translation

The transformation from Exercise 7 is $x \mapsto R_{60°}x + (3+4\sqrt{3}, 4-3\sqrt{3})^T$. Hence it is rotation about zero followed by translation by

$$\boxed{p = (3+4\sqrt{3},\ 4-3\sqrt{3})^T}.$$

The translation is the original center minus its rotated image; it is not simply the center $(6,8)$ itself.

Exercise 2.7.15: Normalize a homogeneous four-vector

A homogeneous point (X,Y,Z,w) with $w \neq 0$ represents $(X/w, Y/w, Z/w)$. Dividing by $1/24$ multiplies the first three entries by 24, so the represented vector is $\boxed{(12,-6,-3)^T}$ in $\mathbb{R}^3$.

Exercise 2.7.16: Different representatives of one point

Yes: $(10,-20,-30,40) = 10(1,-2,-3,4)$. Dividing either representative by its final coordinate gives $(1/4,-1/2,-3/4)$. A common nonzero scale changes the homogeneous representative but not the point it represents.

Exercise 2.7.17: A three-dimensional rotation about the x-axis

The x-coordinate is unchanged, while the (y, z) coordinates undergo the usual 60° planar rotation in the orientation shown. The homogeneous matrix is

$$\boxed{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2 & -\sqrt{3}/2 & 0 \\ 0 & \sqrt{3}/2 & 1/2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}.$$

It sends e_2 toward e_3, confirming the positive orientation about e_1 by the right-hand convention.

Exercise 2.7.18: Rotate about z, then translate

The rotation leaves z fixed and has planar angle -30°. Translation after rotation appends $(5, -2, 1)^T$ as the last column:

$$\boxed{\begin{bmatrix} \sqrt{3}/2 & 1/2 & 0 & 5 \\ -1/2 & \sqrt{3}/2 & 0 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}}.$$

For example the origin is sent to $(5, -2, 1)$, checking the order of actions.

Exercise 2.7.19: Project a triangle from a finite viewing point

A line from $c = (0, 0, 10)$ through $v = (x, y, z)$ is $c + t(v - c) = (tx, ty, 10 + t(z - 10))$. Its intersection with the image plane $z = 0$ has $t = 10/(10 - z)$. Consequently the projection formula is

$$(x, y, z) \longmapsto \left(\frac{10x}{10 - z}, \frac{10y}{10 - z}, 0 \right).$$

For the three given vertices the scale factors are $5/3, 5/4, 5/2$, so their images are

$$\boxed{(7, 2, 0), \quad (15/2, 5, 0), \quad (5, 5, 0)}.$$

All original points have $z < 10$, so each line meets the image plane, and the triangular region projects to the triangle with these vertices.

Exercise 2.7.20: A second perspective triangle

Apply the same formula. At $(7, 3, -5)$ the scale factor is $2/3$; at $(12, 8, 2)$ it is $5/4$; at $(1, 2, 1)$ it is $10/9$. Thus the image triangle has vertices

$$\boxed{(14/3, 2, 0), \quad (15, 10, 0), \quad (10/9, 20/9, 0)}.$$

Perspective projection is not a single ordinary 3×3 linear map on the original three coordinates: the required division depends on z. Homogeneous coordinates encode that division by a final normalization.

Exercise 2.7.21: Convert standard color data into screen data

Let M denote the stated CIE conversion matrix, so $M(R,G,B)^T = (X,Y,Z)^T$. To recover the gun data solve this system:

$$\boxed{\begin{bmatrix} R \\ G \\ B \end{bmatrix} = M^{-1} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}}, \qquad M^{-1} = \frac{1}{20224} \begin{bmatrix} 45677 & -21023 & -7023 \\ -27293 & 47407 & 1407 \\ 1840 & -6160 & 25840 \end{bmatrix}.$$

This exact rational inverse treats the printed decimals as exact input. Its decimal approximation is

$$\begin{bmatrix} 2.258554 & -1.039508 & -.347261 \\ -1.349535 & 2.344096 & .069571 \\ .090981 & -.304589 & 1.277690 \end{bmatrix}.$$

Exact matrix multiplication checks $MM^{-1} = I$; in particular the conversion is uniquely reversible. A matrix program can reproduce it with SymPy rational decimal entries and `M.inv()`; do not reverse the conversion by multiplying by M a second time.

Exercise 2.7.22: Recover RGB from a television signal

Write the supplied matrix as N, so the transmitted vector is $(Y,I,Q)^T = N(R,G,B)^T$. Solving for RGB gives

$$\boxed{\begin{bmatrix} R \\ G \\ B \end{bmatrix} = N^{-1} \begin{bmatrix} Y \\ I \\ Q \end{bmatrix}},$$

$$N^{-1} = \frac{1}{50845523} \begin{bmatrix} 51002600 & 48549800 & 31415400 \\ 50681600 & -13764200 & -32784600 \\ 51277600 & -56463200 & 86415400 \end{bmatrix} \approx \begin{bmatrix} 1.003089 & .954849 & .617860 \\ .996776 & -.270706 & -.644788 \\ 1.008498 & -1.110485 & 1.699568 \end{bmatrix}.$$

The exact form was calculated and verified by rational matrix multiplication. It uses the coefficients as printed, including $.212, -.528, .311$ in the last row. Different engineering conventions or more accurate calibration coefficients would produce a different numerical inverse; no such change has been silently substituted here.